

## Seung Wook Kim

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### DEGREES

*Doctor of Philosophy - Machine Learning* June 2024  
Department of Computer Science, University of Toronto  
Thesis: Controllable Scene Generation with Neural Networks  
Advisor: Prof. Sanja Fidler

*Master of Science - Machine Learning* January 2019  
Department of Computer Science, University of Toronto  
Thesis: Visual Reasoning by Progressive Module Networks  
Advisor: Prof. Sanja Fidler

*Honours Bachelor of Science with High Distinction* June 2016  
Department of Computer Science, University of Toronto  
Computer Science Specialist - Focus in Artificial Intelligence  
Cumulative GPA 3.99/4.00

### RESEARCH INTERESTS

Generative models, robot and physical AI, neural content creation, 3D reconstruction and perception, scene understanding, representation learning

### EMPLOYMENT

*Assistant Professor* August 2026 - Present  
Kim Jaechul Graduate School of AI, KAIST

*Senior Research Scientist* June 2023 - July 2026  
NVIDIA, Spatial Intelligence Lab

*Research Scientist* January 2020 - May 2023  
NVIDIA, Spatial Intelligence Lab

*Research Intern* January 2019 - December 2019  
NVIDIA, Spatial Intelligence Lab

*Research Intern* May 2018 - August 2018  
SKTelecom, T-brain

*Research Scientist* July 2016 - August 2017  
Lunit Inc.

*Part-Time Research Engineer* September 2015 - April 2016  
Avenir

*Software Developer Co-op*, Full-Stack Rails developer May 2014 - August 2015  
IBM Canada

### JOURNAL PUBLICATIONS

*Connecting the Objects: Relational Occlusion Graphs for Amodal Segmentation*

Pattern Recognition Letters, 2026.  
Liu, H., Cho, E., Choi, S., Jung, A., **Kim, S.W.**, Hong, S.

*Self-supervised driven consistency training for annotation efficient histopathology image analysis*  
Medical Image Analysis, 2022.  
Srinidhi, C., **Kim, S.W.**, Chen, F., Martel, A.

## CONFERENCE PUBLICATIONS

\* denotes equal contribution

*Direct Diffusion Score Preference Optimization via Stepwise Contrastive Policy-Pair Supervision*  
European Conference on Computer Vision (ECCV) 2026.  
Kim, D., Lyu, S., **Kim, S.W.**, Seo, P.H.

*Random Conditioning with Distillation for Data-Efficient Diffusion Model Compression*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2025.  
Kim, D.\*, Park, S.\*, Han, G., **Kim, S.W.**, Seo, P.H.

*L4GM: Large 4D Gaussian Reconstruction Model*  
Conference on Neural Information Processing Systems (NeurIPS) 2024.  
Ren, J., Xie, K., Mizraei, A., Liang, H., Zeng, X., Kreis, K., Liu, Z., Torralba, A., Fidler, S., **Kim, S.W.**, Ling, H.

*DistillNeRF: Perceiving 3D Scenes from Single-Glance Images by Distilling Neural Fields and Foundation Model Features*  
Conference on Neural Information Processing Systems (NeurIPS) 2024.  
Wang, L., **Kim, S.W.**, Yang, J., Yu, C., Ivanovic, B., Waslander, S., Wang, Y., Fidler, S., Pavone, M., Karkus, P.

*Diffusion Texture Painting*  
SIGGRAPH 2024.  
Hu, A., Desai, N., Alhaija, H., **Kim, S.W.**, Shugrina, M.

*Align Your Gaussians: Text-to-4D with Dynamic 3D Gaussians and Composed Diffusion Models (Highlight)*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2024.  
Ling, H.\*, **Kim, S.W.\***, Torralba, A., Fidler, S., Kreis, K.

*EmerDiff: Emerging Pixel-level Semantic Knowledge in Diffusion Models*  
International Conference on Learning Representations (ICLR) 2024.  
Namekata, K., Sabour, A., Fidler, S., **Kim, S.W.**

*EmerNeRF: Emergent Spatial-Temporal Scene Decomposition via Self-Supervision*  
International Conference on Learning Representations (ICLR) 2024.  
Yang, J., Ivanovic, B., Litany, O., Weng, X., **Kim, S.W.**, Li, B., Che, T., Xu, D., Fidler, S., Pavone, M., Wang, Y.

*WildFusion: Learning 3D-Aware Latent Diffusion Models in View Space*  
International Conference on Learning Representations (ICLR) 2024.  
Schwarz, K., **Kim, S.W.**, Gao, J., Fidler, S., Geiger, A., Kreis, K.

*DreamTeacher: Pretraining Image Backbones with Deep Generative Models*

International Conference on Computer Vision (ICCV) 2023.  
Li, D.\*, Ling, H.\*, Kar, A., Acuna, D., **Kim, S.W.**, Kreis, K., Torralba, A., Fidler, S.

*NeuralField-LDM: Scene Generation with Hierarchical Latent Diffusion Models*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2023.  
**Kim, S.W.\***, Brown, B.\*, Yin, K., Kreis, K., Schwarz, K., Li, D., Rombach, R.,  
Torralba, A., Fidler, S.

*Align your Latents: High-Resolution Video Synthesis with Latent Diffusion Models*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2023.  
Blattmann, A.\*, Rombach, R.\*, Ling, H.\*, Dockhorn, T.\*, **Kim, S.W.**, Fidler, S.,  
Kreis, K.

*PolymorphicGAN: Generating Aligned Samples Across Multiple Domains With Learned Morph Maps (Oral)*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2022.  
**Kim, S.W.**, Kreis, K., Li, D., Torralba, A., Fidler, S.

*BigDatasetGAN: Synthesizing ImageNet with Pixel-wise Annotations*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2022.  
Li, D., Ling, H., **Kim, S.W.**, Kreis, K., Barriuso, A., Fidler, S., Torralba, A.

*EditGAN: High-Precision Semantic Image Editing*  
Conference on Neural Information Processing Systems (NeurIPS) 2021.  
Ling, H., Kreis, K., Li, D., **Kim, S.W.**, Torralba, A., Fidler, S.

*DriveGAN: Towards a Controllable High-Quality Neural Simulation (Oral)*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2021.  
**Kim, S.W.**, Phillion, J., Torralba, A., Fidler, S.

*Variational Amodal Object Completion*  
Conference on Neural Information Processing Systems (NeurIPS) 2020.  
Ling, H., Acuna, D., Kreis, K., **Kim, S.W.**, Fidler, S.

*Learning to Simulate Dynamic Environments with GameGAN*  
Conference on Computer Vision and Pattern Recognition (CVPR) 2020.  
**Kim, S.W.**, Zhou, H., Phillion, J., Torralba, A., Fidler, S.

*Visual Reasoning by Progressive Module Networks*  
International Conference on Learning Representations (ICLR) 2019.  
**Kim, S.W.**, Tapaswi, M., Fidler, S.

*Keep and Learn: Continual Learning by Constraining the Latent Space for Knowledge Preservation in Neural Networks*  
Medical Image Computing and Computer Assisted Intervention (MICCAI) 2018.  
Kim, H.E., **Kim, S.W.**, Lee, J.

## TECHNICAL REPORTS

*Instant NuRec: Feed-Forward 3D Gaussian Reconstruction for Driving Scene Simulation*  
ArXiv Preprint, 2026.  
NVIDIA.

*NVIDIA OmniDreams: Real-Time Generative World Model for Closed-Loop Autonomous Vehicle Simulation*

ArXiv Preprint, 2026.  
NVIDIA.

*Cosmos-Drive-Dreams: Scalable Synthetic Driving Data Generation with World Foundation Models*

ArXiv Preprint, 2025.

Ren, X.\*, Lu, Y\*, Cao, T.\*, Gao, R.\*, Huang, S., Sabour, A., Shen, T., Pfaff, T., Wu, J., Chen, R., **Kim, S.W.**, Gao, J., Leal-Taixe, L., Chen, M., Fidler, S., Ling, H.

*Cosmos World Foundation Model Platform for Physical AI*

ArXiv Preprint, 2025.

NVIDIA.

*The Shmoop Corpus: A Dataset of Stories with Loosely Aligned Summaries*

ArXiv Preprint, 2019.

Chaudhury, A., Tapaswi, M., **Kim, S.W.**, Fidler, S.

## WORKSHOP PUBLICATIONS

*Cascaded Pyramid Network for 3D Human Pose Estimation Challenge*

European Conference on Computer Vision (ECCV) 2018.

Hong, S., Jung, W., Woo, I., **Kim, S.W.**

*Transferring Knowledge To Smaller Network With Class-Distance Loss*

International Conference on Learning Representations (ICLR) workshop 2017.

**Kim, S.W.**, Kim, H.E.

*Combining word prediction and r-ary Huffman coding for text entry*

InterSpeech Speech and Language Processing for Assistive Technologies (SLPAT) 2016.

**Kim, S.W.**, Rudzicz, F.

## AWARDS & SCHOLARSHIPS

*DiDi Graduate Student Award*

2020

Awarded to graduate students based on their academic standing and research potential  
,University of Toronto

*Principal Janet Paterson Award*

2016

Awarded to the student graduating with the highest grade point average from Innis college, University of Toronto

*Dean's List*

2013-2016

Recognition of exceptional academic achievement

*Samuel Beatty In-Course Award*

2014

Awarded to students in the Departments of Mathematics, Physics, Statistics or Computer Science for outstanding academic performance

*NSERC USRA*

2013

Undergraduate Student Research Awards

*University of Toronto In-Course Scholarship*

2013

Awarded to students who demonstrate academic merit

*Later Life Learning Scholarship*

2013

Awarded to students in the Faculty of Arts and Science for outstanding academic performance

*AUCC Bunge Canada Scholarship* 2011-2013  
Awarded for high academic achievement

*Governor General's Bronze Medal* 2009  
Awarded to the student graduating with the highest grade point average from a Canadian high school